

Projects

00103335: Deep Learning and Reinforcement Learning

Fall 2026

Objective. Use analysis and simulation to study four phenomena: double descent in ridgeless least squares, uncertainty quantification for averaged stochastic gradient descent (SGD), first-order optimizer trajectories on a nonconvex loss landscape, and the effect of Newton–Schulz coefficients in Muon for matrix-valued optimization. Your submission must be reproducible: state all random seeds, use double precision, and label every axis and curve.

Problem 1. Double-descent experiment

Generate data from

$$y = X\beta + \varepsilon, \quad X_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad \varepsilon \sim \mathcal{N}(0, I_n), \quad n = 200, \quad \beta = \sqrt{5} e_1,$$

and vary the number of features over

$$p \in \{20, 40, 60, 80, 100, 120, 140, 160, 180, 199, 200, 220, 240, 260, 300, 400, 600\}.$$

For each value of p , fit the minimum-norm least-squares estimator $\hat{\beta} = X^+y$. Generate an independent test matrix $X_{\text{test}} \in \mathbb{R}^{2000 \times p}$ with i.i.d. standard Gaussian entries, and estimate the prediction risk by

$$R = \frac{1}{2000} \left\| X_{\text{test}} \hat{\beta} - X_{\text{test}} \beta \right\|_2^2.$$

- (a) Repeat the experiment 20 times independently for every p , recording R in each repetition.
- (b) Make two separate plots of R against p : one showing the median and one showing the sample mean. Use each plot to identify the value of p at which the peak occurs.
- (c) Repeat the same experiment, including both plots, for $\beta = \sqrt{2} e_1$ and $\beta = e_1$. Use the same values of p and the same number of repetitions.

Problem 2. Asymptotically valid inference for averaged SGD

Let

$$\sigma(u) = \frac{1}{1 + e^{-u}}, \quad \Sigma_X = \begin{pmatrix} 1 & 0.4 \\ 0.4 & 1 \end{pmatrix}, \quad \theta^* = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

At iteration t , draw a fresh observation, independently over time,

$$X_t \sim \mathcal{N}_2(0, \Sigma_X), \quad Y_t \mid X_t \sim \text{Bernoulli}(\sigma(X_t^\top \theta^*)).$$

For $z = (x, y)$, use the two-dimensional logistic loss

$$\ell(\theta; z) = \log(1 + e^{x^\top \theta}) - y x^\top \theta, \quad L(\theta) = \mathbb{E}\{\ell(\theta; Z)\}.$$

The inferential target is the scalar contrast

$$\psi^* = c^\top \theta^* = -\frac{1}{\sqrt{5}}, \quad c = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

- (a) Show that the stochastic gradient and per-observation Hessian are

$$g(\theta; x, y) = \{\sigma(x^\top \theta) - y\}x, \quad A(\theta; x) = \sigma(x^\top \theta)\{1 - \sigma(x^\top \theta)\}xx^\top.$$

Explain why θ^* is the unique minimizer of L .

- (b) Starting from $\theta_0 = (0, 0)^\top$, run the one-pass recursion

$$\theta_t = \theta_{t-1} - \eta_t g(\theta_{t-1}; X_t, Y_t), \quad \eta_t = 1.6(t + 10)^{-0.6}.$$

Fix the burn-in at $t_0 = 200$. For $T > t_0$, define

$$I_T = \{t_0 + 1, \dots, T\}, \quad n_T = T - t_0, \quad \bar{\theta}_T = \frac{1}{n_T} \sum_{t \in I_T} \theta_t, \quad \hat{\psi}_T = c^\top \bar{\theta}_T.$$

Use fresh data at every step; do not cycle through a fixed data set. Using this averaged estimator, construct each of the following nominal 95% confidence intervals.

Plug-in. With $g_t = g(\theta_{t-1}; X_t, Y_t)$, compute

$$\hat{H}_T = \frac{1}{n_T} \sum_{t \in I_T} A(\theta_{t-1}; X_t), \quad \hat{S}_T = \frac{1}{n_T} \sum_{t \in I_T} g_t g_t^\top, \quad \hat{V}_T = \hat{H}_T^{-1} \hat{S}_T \hat{H}_T^{-1}.$$

Use the full sandwich form, even though the information identity holds in the correctly specified population model. The interval is

$$\mathcal{I}_T^{\text{PI}} = \left[\hat{\psi}_T \pm 1.96 \sqrt{\frac{c^\top \hat{V}_T c}{n_T}} \right].$$

Online multiplier bootstrap. Maintain $B = 200$ perturbed paths. Independently draw $U_t^{(b)} \stackrel{\text{iid}}{\sim} \text{Exp}(1)$, independently of the data, and update

$$\theta_0^{*(b)} = \theta_0, \quad \theta_t^{*(b)} = \theta_{t-1}^{*(b)} - \eta_t U_t^{(b)} g(\theta_{t-1}^{*(b)}; X_t, Y_t), \quad b = 1, \dots, B.$$

All bootstrap paths in an outer replication must use the same observed stream $\{(X_t, Y_t)\}$; only their multipliers differ. Define

$$\bar{\theta}_T^{*(b)} = \frac{1}{n_T} \sum_{t \in I_T} \theta_t^{*(b)}, \quad D_T^{*(b)} = \sqrt{n_T} c^\top (\bar{\theta}_T^{*(b)} - \bar{\theta}_T).$$

If $q_{p,T}^*$ is the empirical p -quantile of the B values $D_T^{*(b)}$, use the basic bootstrap interval

$$\mathcal{I}_T^{\text{Boot}} = \left[\hat{\psi}_T - \frac{q_{0.975,T}^*}{\sqrt{n_T}}, \hat{\psi}_T - \frac{q_{0.025,T}^*}{\sqrt{n_T}} \right].$$

Random scaling. Relabel the retained iterates as $\vartheta_j = \theta_{t_0+j}$, $j = 1, \dots, n_T$, and let

$$\bar{\vartheta}_s = \frac{1}{s} \sum_{j=1}^s \vartheta_j, \quad \hat{\psi}_s = c^\top \bar{\vartheta}_s, \quad \hat{r}_T = \frac{1}{n_T^2} \sum_{s=1}^{n_T} s^2 (\hat{\psi}_s - \hat{\psi}_T)^2.$$

Use

$$\mathcal{I}_T^{\text{RS}} = \left[\hat{\psi}_T \pm 6.747 \sqrt{\frac{\hat{r}_T}{n_T}} \right].$$

The value 6.747 is the 0.975 critical value of the scalar random-scaling limit distribution.

(c) Repeat the complete experiment independently $R = 500$ times, using

$$T \in \{500, 1000, 2000, 5000, 10000\}.$$

Within one outer replication, run a single data stream and its B bootstrap paths to $T = 10000$, recording all five checkpoints. For $m \in \{\text{PI}, \text{Boot}, \text{RS}\}$, estimate the empirical coverage rate

$$\hat{C}_m(T) = \frac{1}{500} \sum_{r=1}^{500} \mathbf{1}\{\psi^* \in \mathcal{I}_{T,r}^m\}.$$

Make one figure whose horizontal axis is the number of iterations T and whose vertical axis is empirical coverage rate. Plot one labeled curve per method, a horizontal line at 0.95, and the Monte Carlo reference band

$$0.95 \pm 1.96 \sqrt{\frac{0.95(0.05)}{500}} \approx [0.931, 0.969].$$

Also report the mean interval length for each method at $T = 10000$.

Problem 3. Optimizer trajectories on a nonconvex loss

For $w = (x, y)^\top \in \mathbb{R}^2$, consider

$$F(x, y) = \frac{1}{4}(x^2 - 1)^2 + 2\left(y - \frac{x^2}{2}\right)^2.$$

- (a) Derive the gradient and Hessian of F . Find and classify all stationary points, and determine which, if any, are global minimizers. Justify your conclusions.
- (b) For every stochastic optimizer, use the noisy gradient

$$g_k = \nabla F(w_k) + \sigma z_k, \quad z_k \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_2), \quad \sigma = 0.1.$$

Run $K = 1000$ updates from each of the following initial points:

$$w_0^{(1)} = (0, 0), \quad w_0^{(2)} = (0, 1.5), \quad w_0^{(3)} = (-1.5, 2.0), \quad w_0^{(4)} = (0.8, -0.5).$$

GD uses the exact gradient, while the other five optimizers use the noisy gradient g_k above. Use the following parameter choices as baselines:

optimizer	parameters
GD	learning rate 0.03
SGD	learning rate 0.03
Momentum	learning rate 0.008, momentum 0.9
AdaGrad	learning rate 0.15, $\varepsilon = 10^{-8}$
RMSProp	learning rate 0.03, $\rho = 0.9$, $\varepsilon = 10^{-8}$
Adam	learning rate 0.03, $(\beta_1, \beta_2) = (0.9, 0.999)$, $\varepsilon = 10^{-8}$

You are encouraged to experiment with alternative learning rates and decay parameters; report all parameter values used. Visualize the trajectories. For each initial point, make one figure showing the trajectories of all six optimizers on the same axes.

Problem 4. Muon with alternative Newton–Schulz coefficient schedules

Let $n = 128$, $d = 64$, and $q = 32$, and consider the matrix-valued least-squares objective

$$F(W) = \frac{1}{2n} \|XW - Y\|_F^2, \quad W \in \mathbb{R}^{d \times q}.$$

For a full-column-rank matrix Z , define $\text{qf}(Z)$ by computing a reduced QR factorization $Z = QR$, setting $D = \text{diag}\{\text{sign}(R_{11}), \dots, \text{sign}(R_{rr})\}$, and taking $\text{qf}(Z) = QD$, where r is the number of columns and $\text{sign}(0) = 1$. Draw, in the listed order, three independent standard-Gaussian matrices

$$Z_1 \in \mathbb{R}^{n \times d}, \quad Z_2 \in \mathbb{R}^{d \times d}, \quad Z_3 \in \mathbb{R}^{q \times q},$$

and set

$$Q_X = \text{qf}(Z_1), \quad V_X = \text{qf}(Z_2), \quad V_\star = \text{qf}(Z_3).$$

For $j = 1, \dots, d$, let

$$\lambda_j = \begin{cases} 10^{-4+4(j-1)/(q-1)}, & 1 \leq j \leq q, \\ 10^{-4}, & q < j \leq d, \end{cases}$$

and construct

$$X = \sqrt{n} Q_X \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_d}) V_X^\top, \quad W^\star = V_X(:, 1:q) V_\star^\top, \quad Y = XW^\star.$$

Here $V_X(:, 1:q)$ denotes the first q columns of V_X .

- Let $H = n^{-1} X^\top X$. Derive $\nabla F(W)$ in terms of H and $W - W^\star$, determine the eigendecomposition of H , and prove that $W^\star$ is the unique minimizer.
- A coefficient choice in this problem is a five-step schedule

$$\mathcal{C}^{(r)} = ((a_0^{(r)}, b_0^{(r)}, c_0^{(r)}), \dots, (a_4^{(r)}, b_4^{(r)}, c_4^{(r)})).$$

Define its scalar maps and their five-step composition by

$$p_k^{(r)}(s) = a_k^{(r)} s + b_k^{(r)} s^3 + c_k^{(r)} s^5, \quad \Psi_r = p_4^{(r)} \circ p_3^{(r)} \circ \dots \circ p_0^{(r)}.$$

Compare the following six *discrete* choices. The first four reuse one triple in every step:

r	schedule name	$(a_k^{(r)}, b_k^{(r)}, c_k^{(r)}), \quad k = 0, \dots, 4$
C3	classical cubic	$(3/2, -1/2, 0)$
T5	fifth-order Taylor	$(15/8, -5/4, 3/8)$
KJ	original Muon	$(3.4445, -4.7750, 2.0315)$
S5	Su shared fit	$(3.3748, -4.6969, 2.1433)$

The SX3 schedule uses a different row at each step:

k	$a_k^{(\text{SX3})}$	$b_k^{(\text{SX3})}$	$c_k^{(\text{SX3})}$
0	4.6182	-12.9582	9.3299
1	3.8496	-7.9585	4.3052
2	3.5204	-7.2918	4.0606
3	3.2067	-6.8243	4.2802
4	3.2978	-5.7848	3.8917

Finally, the five-step PE5 schedule uses

$$(a_k^{(\text{PE5})}, b_k^{(\text{PE5})}, c_k^{(\text{PE5})}) = \left(\frac{\bar{a}_k}{1.01}, \frac{\bar{b}_k}{1.01^3}, \frac{\bar{c}_k}{1.01^5} \right),$$

where

k	$\bar{a}_k$	$\bar{b}_k$	$\bar{c}_k$
0	8.28721	-23.5959	17.3004
1	4.10706	-2.94785	0.544843
2	3.94869	-2.90890	0.551819
3	3.31842	-2.48849	0.510049
4	2.30065	-1.66890	0.418807

Over $s \in [10^{-4}, 1]$, evaluate all six schedules on a 1,001-point logarithmic grid. In a single log-log figure, plot $|\Psi_r(s)|$ for all six schedules. Compare the schedules and summarize the differences you observe.

- (c) Implement $\text{NS5}_r(B)$. For a nonzero matrix $B \in \mathbb{R}^{d \times q}$, initialize

$$Z_0 = \frac{B^\top}{\|B^\top\|_F + \varepsilon_{\text{NS}}}, \quad \varepsilon_{\text{NS}} = 10^{-7}.$$

For $k = 0, \dots, 4$, read row k of schedule r and compute

$$A_k = Z_k Z_k^\top, \quad Z_{k+1} = a_k^{(r)} Z_k + (b_k^{(r)} A_k + c_k^{(r)} A_k^2) Z_k.$$

Return $\text{NS5}_r(B) = Z_5^\top$.

Validate every schedule on

$$B_{\text{test}} = V_X(:, 1:q) \text{diag}(\tau_1, \dots, \tau_q) V_\star^\top, \quad \tau_j = 10^{-3(j-1)/(q-1)}.$$

For each schedule, compute the relative polar error

$$E_{\text{polar}}(r) = \frac{\|\text{NS5}_r(B_{\text{test}}) - V_X(:, 1:q) V_\star^\top\|_F}{\sqrt{q}}.$$

- (d) For each schedule r , start a separate run from $W_0^{(r)} = 0$ and $B_0^{(r)} = 0$. For $t = 0, \dots, K-1$, with $K = 200$, use

$$\begin{aligned} G_t^{(r)} &= \nabla F(W_t^{(r)}), \\ B_{t+1}^{(r)} &= 0.9 B_t^{(r)} + 0.1 G_t^{(r)}, \\ \tilde{O}_t^{(r)} &= \text{NS5}_r(B_{t+1}^{(r)}), \\ W_{t+1}^{(r)} &= W_t^{(r)} - \eta_t \tilde{O}_t^{(r)}, \quad \eta_t = 0.03. \end{aligned}$$

Produce a two-panel figure with logarithmic vertical axes showing

$$\frac{F(W_t^{(r)})}{F(W_0)} \quad \text{and} \quad \frac{\|W_t^{(r)} - W^\star\|_F}{\|W^\star\|_F}.$$

Compare the six schedules and explain the principal differences you observe.

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